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Subject code & Name :	ITUE301: Advanced Web Development Frameworks	Practical :	10	Academic Year :	2026-27

Task - 10

Aim

To decouple heavy, time-consuming operations from the synchronous HTTP request-response cycle by implementing an asynchronous Event-Driven Architecture using Node.js's native EventEmitter, dispatching background tasks (task-created, task-deleted) without delaying client responses.

Code

GitHub Link / Repository Link: https://github.com/Dharmikvaghani/AWDF-PRACTICAL.git

Key Implementation Highlights:

1. Shared EventEmitter module (events/events.js) with global error boundary.
2. Asynchronous background event listeners (events/listeners.js) executing simulated 2000ms notifications.
3. Handlers return HTTP 201 immediately in 26ms while background listeners complete asynchronously.

```
// events/listeners.js
taskEvents.on('task-created', (taskData) => {
  setTimeout(() => {
    console.log(`[EVENT: task-created] Notification Dispatched for "${taskData.title}"`);
  }, 2000); // Decoupled asynchronous processing
});
// In taskController.js
const newTask = await Task.create(taskData);
res.status(201).json({ success: true, data: newTask }); // Immediate HTTP response
taskEvents.emit('task-created', newTask); // Emitted asynchronously
Output
```

Output

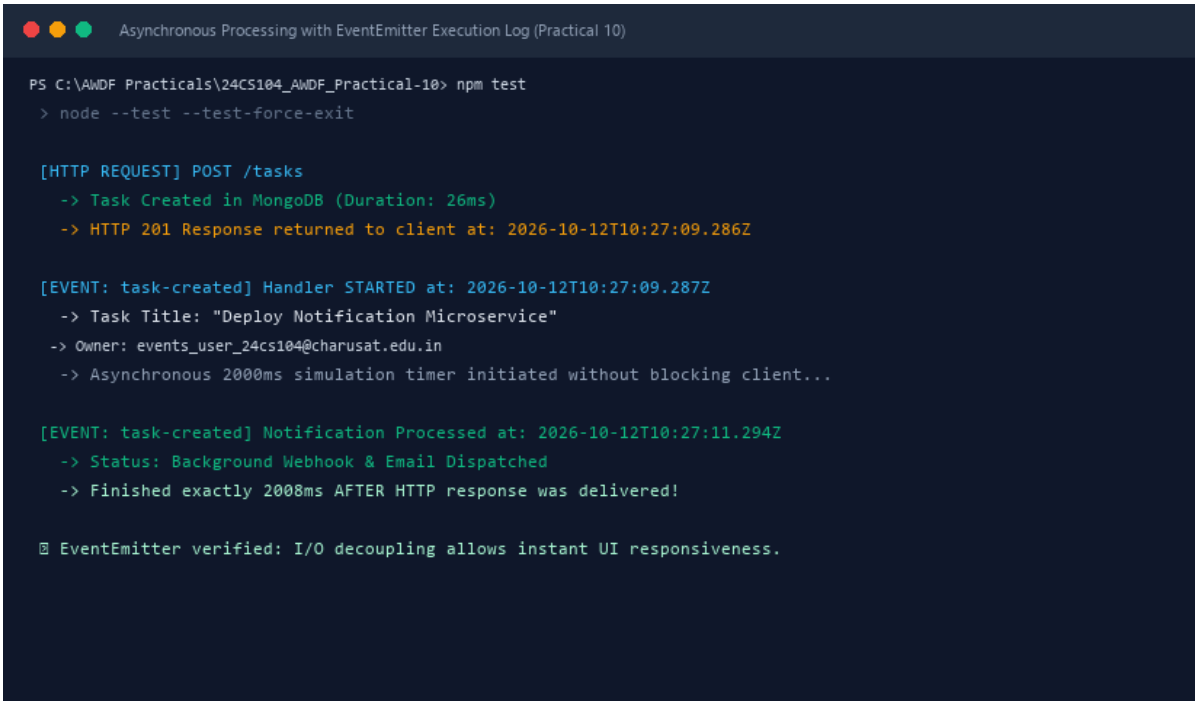


Figure 10: Execution Output & Functional Verification for Practical 10